Introductory Chemistry

A step-by-step study plan: from the most basic ideas to the essential core of the subject

The big picture. Chemistry builds in layers. This plan follows one guiding question at each stage: **What is matter?** → **What is it made of?** → **How do its particles bond?** → **How does it change?** → **How much does it change?** Each unit relies on the ones before it, so work through them in order. Every unit ends with a box of hand-picked videos and interactive simulations (all links are clickable).

Phase	Units	Central question	Suggested time
1. Foundations	1–3	What is matter and what are atoms like?	5–6 weeks
2. Organization & bonding	4–6	How are elements organized and how do they combine?	6–7 weeks
3. Change & quantity	7–9	How do substances react, and in what amounts?	8–9 weeks
4. Reactivity	10	Why do some reactions happen and others not?	3–4 weeks

Times assume roughly 3–4 hours of study per week. Adjust to your pace.

PHASE 1 · Foundations: matter and measurement

Unit 1 | Matter and Measurement (~2 weeks)

Learning goal: Describe, classify and measure matter using the language of science.

Core topics

- What matter is: mass, volume and density
- States of matter and phase changes (melting, boiling, sublimation, etc.)
- Physical vs. chemical properties; physical vs. chemical changes
- Pure substances (elements, compounds) vs. mixtures (homogeneous, heterogeneous)
- Separation techniques: filtration, distillation, chromatography, evaporation
- SI units, scientific notation, significant figures, unit conversion (dimensional analysis)

You should be able to

» Classify any sample as an element, compound or mixture
» Tell a physical change from a chemical change, with evidence
» Convert between units and calculate density

Why it comes first: without measurement and a way to classify matter, nothing that follows has a foundation. Dimensional analysis, in particular, is the main tool you will use in Unit 7.

Resources
- **Watch:** Crash Course Chemistry · Tyler DeWitt
- **Simulate:** PhET — States of Matter · Density

Unit 2 | The Atom (~1.5 weeks)

Learning goal: Explain what atoms are made of and how our model of the atom developed.

Core topics

- Atomic models over time: Dalton → Thomson → Rutherford → Bohr → quantum model
- Subatomic particles: proton, neutron, electron (charge, mass, location)
- Atomic number (Z), mass number (A), isotopes, average atomic mass
- Ions: cations and anions

You should be able to

» Find the number of protons, neutrons and electrons in any atom or ion
» Calculate average atomic mass from isotope abundances
» Explain which experiment led to each change in the atomic model

Resources
- **Watch:** Professor Dave Explains · Crash Course Chemistry
- **Simulate:** PhET — Build an Atom · Isotopes and Atomic Mass · Rutherford Scattering

Unit 3 | Electron Configuration (~2 weeks)

Learning goal: Describe how electrons are arranged in an atom, and why that arrangement matters.

Core topics

- Energy levels, sublevels (s, p, d, f) and orbitals
- Aufbau principle, Hund's rule, Pauli exclusion principle
- Valence electrons and Lewis dot symbols for atoms
- Noble-gas (condensed) notation

You should be able to

» Write the electron configuration of elements up to Z = 36
» Identify valence electrons and draw Lewis dot symbols
» Predict the most likely ion an element will form

Key unit: electron configuration explains *why* atoms behave the way they do. It is the hinge of the whole course: the periodic table, bonding and reactivity all depend on it.

Resources
- **Watch:** Tyler DeWitt · Professor Dave Explains
- **Extra:** Ptable

PHASE 2 · Organization and bonding

Unit 4 | The Periodic Table (~1.5 weeks)

Learning goal: Use the periodic table as a tool to predict how elements behave.

Core topics

- Structure: groups, periods and blocks (s, p, d, f)
- Metals, nonmetals and metalloids; main families (alkali metals, halogens, noble gases)
- Periodic trends: atomic radius, ionization energy, electronegativity, electron affinity
- How electron configuration determines an element's position

You should be able to

» Locate an element from its configuration, and vice versa
» Compare two elements using periodic trends, and explain the trend
» Predict the general reactivity of a family

Resources
- **Watch:** Periodic Videos · Crash Course Chemistry
- **Extra:** Ptable

Unit 5 | Chemical Bonding (~3 weeks)

Learning goal: Explain how and why atoms join together, and how bond type shapes a substance's properties.

Core topics

- The octet rule (and its exceptions)
- Ionic, covalent (polar, nonpolar, coordinate) and metallic bonds
- Lewis structures of molecules
- Molecular geometry (VSEPR theory) and molecular polarity
- Intermolecular forces: London dispersion, dipole–dipole, hydrogen bonding

You should be able to

» Predict bond type from electronegativity difference
» Draw Lewis structures and determine molecular shape
» Rank substances by boiling point using intermolecular forces

Connection: intermolecular forces explain boiling points and solubility, ideas you will return to in Unit 8 (solutions).

Resources
- **Watch:** Crash Course Chemistry · Tyler DeWitt
- **Simulate:** PhET — Molecule Shapes · Molecule Polarity

Unit 6 | Inorganic Nomenclature (~2 weeks (+ ongoing practice))

Learning goal: Name and write formulas for common inorganic compounds.

Core topics

- Oxidation numbers
- Oxides, hydrides, hydroxides, acids (binary and oxyacids), salts
- IUPAC, Stock and traditional naming systems
- Common polyatomic ions

You should be able to

» Write the formula of a compound from its name, and the name from its formula
» Assign oxidation numbers to every atom in a compound

Study tip: this is the most memorization-heavy unit. Rather than cramming it in one block, practice it a little every week while you move on to the next units.

Resources
- **Watch:** ChemistNATE · Tyler DeWitt

PHASE 3 · Change and quantity

Unit 7 | Chemical Reactions and Stoichiometry (~4 weeks)

Learning goal: Represent chemical changes with equations and calculate the amounts involved.

Core topics

- Chemical equations; law of conservation of mass (Lavoisier)
- Balancing equations (inspection and algebraic methods)
- Reaction types: synthesis, decomposition, single and double displacement, combustion
- **The mole**, Avogadro's number, molar mass
- Stoichiometric calculations: limiting reagent, excess reagent, percent yield
- Percent composition; empirical and molecular formulas

You should be able to

» Balance any common chemical equation
» Convert freely between grams, moles and particles
» Find the limiting reagent and the theoretical and percent yield

The most important unit of the course. Almost all quantitative chemistry depends on mastering the mole. Do not move on until mole conversions feel automatic.

Resources
- **Watch:** Tyler DeWitt · ChemistNATE
- **Simulate:** PhET — Balancing Chemical Equations · Reactants, Products and Leftovers
- **Extra:** ChemCollective Virtual Labs

Unit 8 | Solutions (~2.5 weeks)

Learning goal: Describe how substances dissolve and express how concentrated a solution is.

Core topics

- Solute, solvent, solubility, and factors that affect solubility
- Concentration units: % m/m, % m/v, % v/v, molarity, molality, ppm
- Dilutions ($C_1V_1 = C_2V_2$)
- Colligative properties (introduction)

You should be able to

» Calculate the concentration of a solution in several units
» Plan how to prepare a solution or a dilution in the lab
» Explain solubility using intermolecular forces ("like dissolves like")

Resources
- **Watch:** Khan Academy Chemistry · ChemistNATE
- **Simulate:** PhET — Molarity · Concentration
- **Extra:** ChemCollective Virtual Labs

Unit 9 | Gases (~2 weeks)

Learning goal: Relate pressure, volume, temperature and amount of a gas, and explain gas behavior at the particle level.

Core topics

- Variables: pressure, volume, temperature (in kelvin), amount
- Boyle's, Charles's, Gay-Lussac's and Avogadro's laws
- The ideal gas law (PV = nRT)
- Kinetic molecular theory

You should be able to

» Solve problems with the individual gas laws and the ideal gas law
» Explain each gas law in terms of particle motion

Resources
- **Watch:** Crash Course Chemistry · Professor Dave Explains
- **Simulate:** PhET — Gas Properties

PHASE 4 · Reactivity

Unit 10 | Acids, Bases and Redox (Introduction) (~3–4 weeks)

Learning goal: Understand the two most common families of reactions and get a first look at rates and equilibrium.

Core topics

- Arrhenius and Brønsted–Lowry definitions of acids and bases
- pH and pOH scales; calculations with strong acids and bases
- Neutralization and titration
- Oxidation–reduction: identifying oxidizing and reducing agents, basic balancing
- Introduction to reaction rate and chemical equilibrium (Le Châtelier's principle)

You should be able to

» Calculate the pH of a strong acid or base solution
» Interpret a titration and identify the equivalence point
» Identify what is oxidized and what is reduced in a reaction

Bridge: rates and equilibrium are introduced here only as a foundation. They are developed fully in general chemistry.

Resources
- **Watch:** Crash Course Chemistry · Reactions (ACS)
- **Simulate:** PhET — pH Scale · Acid-Base Solutions · Reactions & Rates
- **Extra:** ChemCollective Virtual Labs

How to study this plan

Critical dependencies

If this unit is weak…	…these units will suffer
Unit 1 (measurement, unit conversion)	Units 7, 8, 9
Unit 3 (electron configuration)	Units 4, 5, 6
Unit 5 (bonding)	Units 6, 8 and organic chemistry
Unit 7 (the mole)	Units 8, 9, 10

- **Solve, don't just read.** Chemistry is learned by working problems. From Unit 7 onward, spend at least 60% of your study time on exercises.
- **Check before moving on.** At the end of each unit, try the "You should be able to" list without notes. If you can't do an item, review it first.
- **Space your practice.** Revisit earlier units briefly every week (especially nomenclature and mole conversions) instead of studying each topic only once.
- **Use videos actively.** Pause before the answer and try the problem yourself. Watching a solution feels like learning, but solving it is what builds the skill.
- **Link it to the lab and daily life.** Cooking, cleaning products, rust, batteries and breathing are all chemistry. Concrete examples make abstract ideas stick.

What's next? Organic chemistry is deliberately left out of this plan. It is a separate block that is best started once covalent bonding and molecular geometry (Unit 5) are solid.

Resource directory

All resources below are free. Video channels are listed by the type of support they offer best.

Video lessons (concept explanations)

- **Crash Course Chemistry** — Fast-paced, animated overview of each topic. Great as a first contact or for review. (youtube.com/@crashcourse)
- **Tyler DeWitt** — Very clear, slow-paced explanations for beginners. Excellent on the mole, stoichiometry and significant figures. (youtube.com/user/tdewitt451)
- **Professor Dave Explains** — Short, structured videos covering the whole general chemistry syllabus, each with a comprehension check. (youtube.com/@ProfessorDaveExplains)
- **Khan Academy Chemistry** — Videos plus graded practice exercises, organized by unit. (khanacademy.org/science/chemistry)

Problem solving

- **ChemistNATE** — Worked examples of typical exercises: balancing, nomenclature, stoichiometry, gases. (youtube.com/user/chemistNATE)

Curiosity and science communication

- **Periodic Videos** — A video for every element of the periodic table, with real demonstrations (University of Nottingham). (youtube.com/user/periodicvideos)
- **Reactions (ACS)** — The chemistry of everyday life: food, medicine, the body (American Chemical Society). (youtube.com/user/ACSReactions)

Interactive tools

- **PhET Interactive Simulations** — Free simulations from the University of Colorado. Run in the browser, no installation. (phet.colorado.edu/en)
- **Ptable** — Interactive periodic table: properties, trends, electron configurations and isotopes. (ptable.com)
- **ChemCollective Virtual Labs** — Virtual lab for practicing solutions, stoichiometry and titrations. (chemcollective.org/vlabs)
- **Chemistry LibreTexts** — Free open-access textbooks to look up any topic in more depth. (chem.libretexts.org)